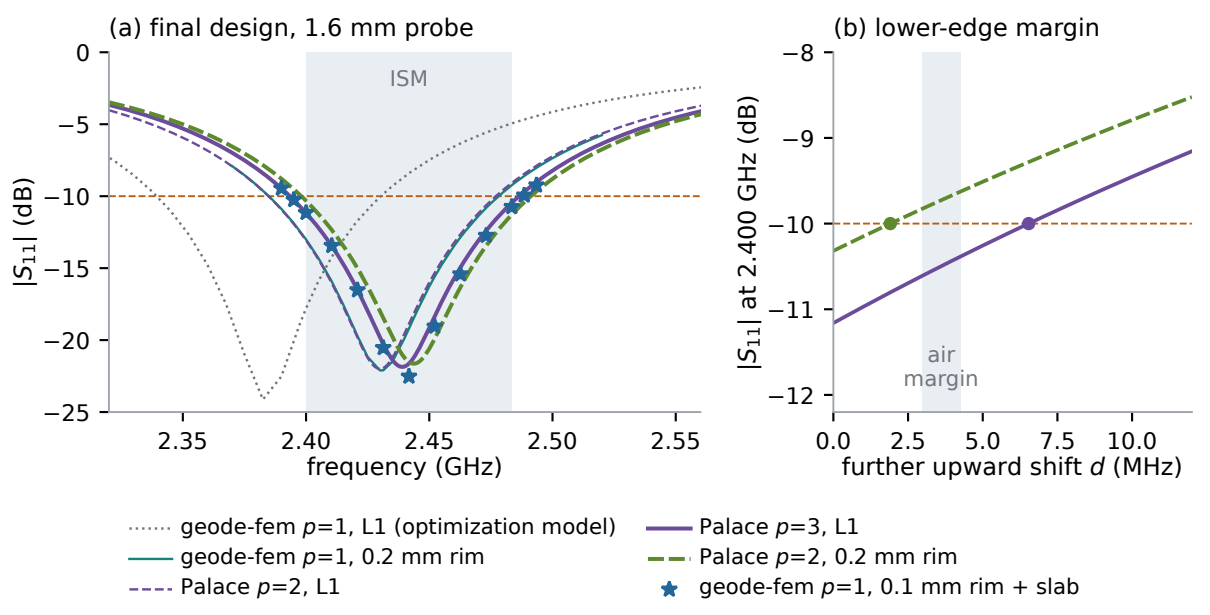
(a) final design, 1.6 mm probe
ISM
$|S_{11}|$ (dB)
0
−5
−10
−15
−20
−25
2.35
2.40
2.45
2.50
2.55
frequency (GHz)
(b) lower-edge margin
air
margin
$|S_{11}|$ at 2.400 GHz (dB)
−8
−9
−10
−11
−12
0.0
2.5
5.0
7.5
10.0
further upward shift d (MHz)
geode-fem p=1, L1 (optimization model)
geode-fem p=1, 0.2 mm rim
Palace p=2, L1
Palace p=3, L1
Palace p=2, 0.2 mm rim
geode-fem p=1, 0.1 mm rim + slab